Kuta Software - Infinite Precalculus

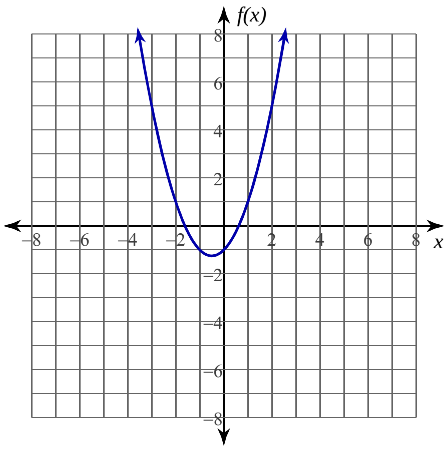
Name____________________

Average Rates of Change

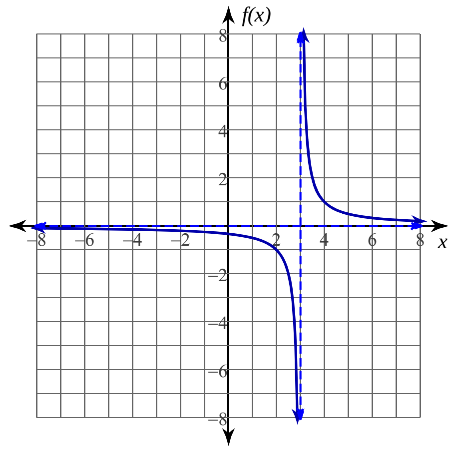
Date________________ Period____

For each problem, find the average rate of change of the function over the given interval.

1) $f(x) = x^2 + x - 1;\quad [-2, -\frac{7}{4}]$



2) $f(x) = \frac{1}{x - 3};\quad [-2, -\frac{3}{2}]$



3) $f(x) = x^2 - x - 1;\quad [3, \frac{13}{4}]$

4) $f(x) = -x^2 - x + 1;\quad [2, \frac{7}{3}]$

5) $f(x) = 2x^2 - 2x - 1;\quad [1, \frac{3}{2}]$

6) $f(x) = \frac{1}{x - 1};\quad [-4, -\frac{11}{3}]$

7) $f(x) = \frac{1}{x + 3};\quad [-1, -\frac{2}{3}]$

8) $f(x) = -\frac{1}{x - 1};\quad [-4, -\frac{7}{2}]$

9) The police have accused a driver of breaking the speed limit of 60 miles per hour. As proof, they provide two photographs. One photo shows the driver's car passing a toll booth at exactly 6 PM. The second photo shows the driver's car passing another toll both 31 miles down the highway at exactly 6:30 PM. Does the photo evidence prove that the driver broke the speed limit during this time?